# Lecture I - Optimal Stopping

Programming: Everyday Decision-Making Algorithms

Dr. Tobias Vlćek

# About this Course

## Teaching Team

(a) **Dr. Tobias Vlcek**

(a) **Dr. Nils Roemer**

## About me

- **Field:** Optimizing and simulating complex systems
- **Languages:** of choice: Julia, Python and Rust
- **Interest:** Modelling, Simulations, Machine Learning
- **Teaching:** OR, Algorithms, and Programming
- **Contact:** vlcek@beyondsimulations.com

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> Tip
>
> We really appreciate active participation and interaction!

## Course Outline

- **I:** Optimal Stopping
- **II:** Explore & Exploit
- **III:** Caching
- **IV:** Scheduling
- **V:** Randomness
- **VI:** Computational Kindness

## Participation

- Try **actively participating** in this course
- You will find it much (!) easier and more fun
- Lecture based on the book Algorithms to live by[1]
- Material and slides are hosted online: beyondsimulations.github.io/Programming-Everyday-Decisions

## Teaching

- **Lecture:** Presentation and discussion of algorithms related to everyday decision-making
- **Tutorial:** Step-by-step assignments to be solved and discussed together in groups
- **Difficulty:** Strongly depends on your background and programming experience

. . .

> Tip
>
> No worries, we will help you out if you have any questions!

## Passing the Course

- **Pass/fail course** without exams
- 75% attendance required for passing the course
- Hand in the assignments of at least **two lectures**
- **Short presentation** and discussion at the end
- You work together in groups of **three students**

## Handing in Assignments

- Each student group submits **one solution**
- Provide us **all** working notebooks of the lecture
- Hand in is due at the **beginning of the next lecture**
- At least **50 %** have to be correct to pass
- You have to pass at least twice

. . .

[1] Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.

> **Tip**
>
> This is just in order to provide you with working solutions after each deadline.

## Learning Python

We will mostly **not cover Python during the lectures!**

. . .

Question: Anybody know why?

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- In our experience, the best way to learn is by **doing**!
- Here, we will focus on decision-making algorithms
- You will learn Python by doing the tutorials

. . .

> **Tip**
>
> Don't worry, we will help you out if you have any questions!

## Difficulty of the Course

- At first it might be a little bit overwhelming
- Programming is similar to learning a **new language**
- First, you have to **get used to it** and learn words
- **Later**, you'll be able to apply it and see results
- Important: *Practice, practice, practice!*

## Goals of the Course

- Learn the basics of programming
- Learn about **algorithmic thinking**
- Be able to **apply** methods and concepts
- **Solve practical problems** with algorithms

> **Tip**
>
> We are convinced that this course will be quite interesting and **teach you more for your daily life** than most other courses!

## Why Python?

- **Origins**: Conceived in late 1980s as a teaching and scripting language
- **Simple Syntax**: Python's syntax is mostly straightforward and very easy to learn
- **Versatility**: Used in web development, data analysis, artificial intelligence, and more
- **Community Support**: A large community of users worldwide and extensive documentation

## Help from AI

- You are allowed to use AI in the course, we use it as well (e.g., Claude, ChatGPT, LLama3 …)
- These **tools are great** for learning Python!
- Can help you a lot **to get started** with programming

. . .

> ⚠ Warning
>
> But you should *not* simply use them to *replace* your learning.

# How to learn programming

## Our Recommendation

1. Be present: Attend the lecture and solve the tutorials
2. Put in some work: Repeat code and try to understand it
3. Do coding: Run code, play around, modify, and solve

. . .

> Tip
>
> Great resources to start are books and small challenges. You can find a list of book recommendations at the end of the lecture. Small challenges to solve can for example be found on Codewars.

## Don't give up!

- Programming is problem solving, don't get **frustrated**!
- Expect to **stretch** your comfort zone

# Setting up Python

## The Setup

- We will use Jupyter Notebooks for the tutorials
- Allow to **combine code and text** in one document
- We will use Visual Studio Code as an IDE

. . .

> Tip
>
> IDE = Integrated Development Environment

## Install Python

- Sources are the Python website or Anaconda
- On macOS, Python is often **already installed**
- If not, I recommend Miniconda (via command line)

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> Tip
>
> If the installation does not work, **let us know!**

## Install VS Code

- Download and install from the website
- Built for **Windows, Linux and Mac**
- Install the **Python** and **Jupyter** extension
- Now you are ready to go!

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> Tip
>
> Unsure on how to work with VS Code and notebooks? Take a look at the tutorial from VS Code and/or ask us! We are happy to help you out!

## Python on iPads

- You **can run Python scripts** on your iPad
- But it is not recommended for the course
- However, you could use Juno **if you want to**
- It works **locally** on your iPad and can run notebooks

. . .

> Caution
>
> Not all packages available in Python are available here, thus you might need a computer to solve certain problems. For our course, this should not be a problem.

# Your first code

## Hello, World!

Task: Create a directory for the course and create a new file called `hello_world.py` with the following code:

```
# This is a comment in Python
print("Hello, World!")
```

```
Hello, World!
```

. . .

Run it with the green 'run' button or by pressing F5!

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> **Note**
>
> "Hello world" is a classic example to start with. Often used as a test to check if your computer is working properly and that you have installed the necessary software.

Any questions

so far?

# Optimal Stopping

## What is Optimal Stopping?

Question: Anybody know what optimal stopping is?

- Optimal stopping is the problem of:
  - choosing the **best option**
  - from a **sequence of options**
  - where the options are revealed **one by one**

Anybody an

example of

optimal stopping?

## Flat Hunting

Photo by Aditya Ghosh on Unsplash

## Hiring applicants

Photo by Scott Graham on Unsplash

## Dating

Photo by Shelby Deeter on Unsplash

## Searching for a parking spot

Photo by Joseph Pearson on Unsplash

# “Secretary Problem”

## The Secretary Problem

- Imagine you’re **hiring a secretary**
- You must interview candidates **one by one**
- Now, you must decide: **hire or continue searching**
- Once you reject a candidate, **you cannot go back**
- How to maximize chance of selecting the best candidate?

. . .

> **Note**
>
> The name is a bit misleading, as the problem is not about hiring a secretary, but about finding the best candidate. It comes from the 1960s and thus a little outdated.

## Basic Setup

- We have `n` candidates
- We interview them **one by one**
- We must decide to **hire or continue searching**
- Ordinal ranking of candidates

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Question: Anybody know what ordinal ranking is?

## Ways to fail

Question: Anybody an idea how we can fail?

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1. Reject all candidates and never hire - stopping too late
2. You hire someone too early - stopping too early

Ideas?

## Look-and-Leap Strategy

The optimal strategy is to:

1. Look at the first 37 % of options
2. Remember the **best one seen so far**
3. Choose the next option that's **better than the best seen**
4. Chance of selecting the best candidate is **37 %**[2]
5. Thus, we can fail with **63 %**!

## Step-by-step Approximation

- With each additional candidate, the chance of getting a new best candidate decreases
  - 1 candidate: 100%
  - 2 candidates: 50%
  - 3 candidates: 33%
  - 4 candidates: 25%
  - 5 candidates: 20%

. . .

Question: Anybody see a pattern?

## Why 37%?

- This is based on the **geometric distribution**
- The optimal stopping point is at `n/e`
- `e` is the base of the natural logarithm (≈ 2.718)
- This comes from maximizing the probability of success

. . .

> **Tip**
>
> This is a bit more advanced, so don't worry if you don't understand it! We will not go into the details of the math here and focus more on the insights.

## Computing the number

```python
import math

percentage = 1/math.e
print(f"Percentage of options to look at: {percentage:.3f}%")

candidates = 20
lookout_phase = candidates/math.e
print(f"Look at first {lookout_phase:.3f} candidates")
```

```
Percentage of options to look at: 0.368%
Look at first 7.358 candidates
```

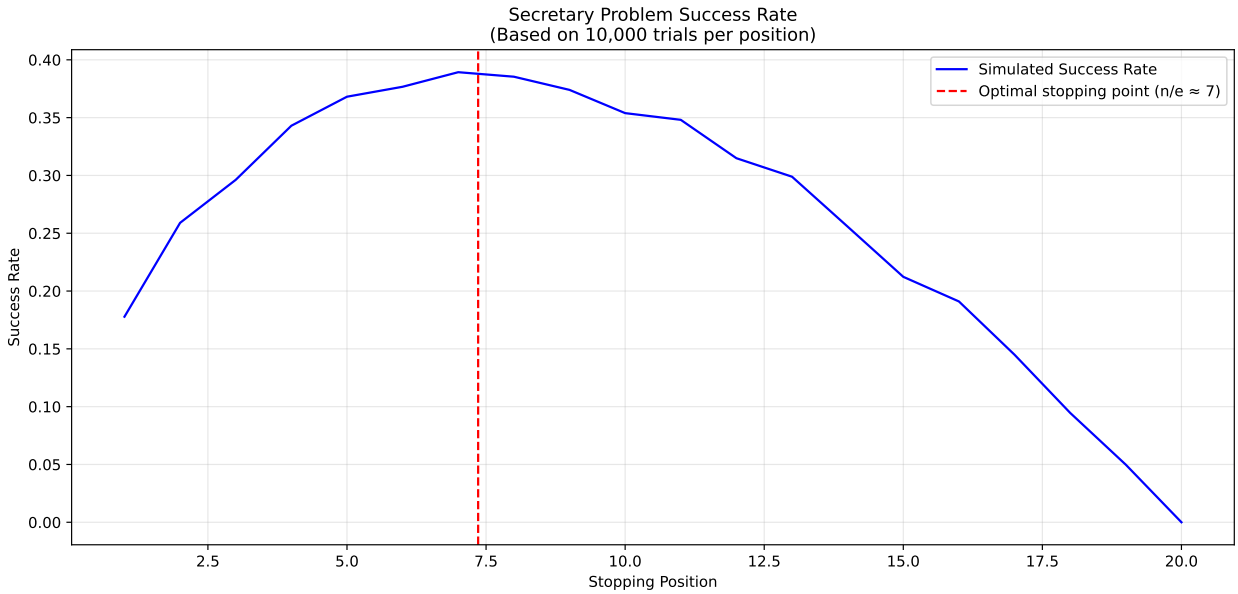
[2] With a large number of candidates. With a small number of candidates, we can do even better.

. . .

> **Note**
>
> No worries if you don't understand the code! We are essentialy just using the formula to calculate the percentage of candidates to look at.

## Geometric Distribution

Let's visualize the success of a simulation with 20 candidates:



# Variations

## Rejection

Question: Imagine a dating scenario, where the other person can also reject you. **Optimal stopping point?**

. . .

- The optimal stopping point is now **lower**
- Because we can now **fail more often**
- With 50 % chance of rejection, we **start leaping at 25 %**

## The role of time

Question: What if we don't have a fixed number of candidates, but rather a fixed amount of time?

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- Imagine we have **one year** to find a new flat
- We want the best flat, but don't know what a good flat is

. . .

- **Same**, but now we decide when to stop searching!

## Other versions

- Selling a house for the **best price** ("Threshold Rule")
- Stealing with a **success probability** ("Burglar's Problem")
- **Finding** a parking spot ("Parking Lot Problem") [3]

[3] An increase in occupancy from 90 to 95% doubles the search time for all drivers!

# Optimal Solution

## The Gambler's Fallacy

Question: Can you imagine a scenario where it would be unwise to use optimal stopping?

. . .

- Imagine a game with a 50 % chance of winning
- If you win, your payoff is a **triple** of your bet
- If you lose, you have to pay your bet

Questions?

## The End

> Note
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> **That's it for todays lecture!**
> We now have covered a brief introduction into optimal stopping and seen how to set up Python. Now we can start with the tutorials!

# Literature

## Interesting Books to start

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.

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> **Note**
>
> The main inspiration for this lecture. Nils and I have read it and discussed it in depth, always wanting to translate it into a course.

## Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Here
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

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> **Note**
>
> Think Python is a great book to start with. It's available online for free. Schrödinger Programmiert Python is a great alternative for German students, as it is a very playful introduction to programming with lots of examples.

## More Literature

For more interesting literature, take a look at the literature list of this course.